Reg. No. [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ]

[18CS5CT02]

FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER–2021

NETWORK SECURITY AND CRYPTOGRAPHY

Time:3 Hours **Maximum Marks:70**

Instructions:
- **i.** Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
- **ii.** Missing data may be suitably assumed.

1.	**a.**	Define Cryptography and Cryptanalysis. Draw and Explain the encryption systems.	**09**
	b.	Explain the difference between symmetric cryptography and asymmetric cryptography. Write a brief not on challenges faced by above two systems.	**05**
2.	**a.**	Explain the conventional model used for information security.	**07**
	b.	Define the terms threat and attack. List and briefly explain the different categories of Security attacks.	**07**
3.	**a.**	What is PlayFair cipher? Explain about PlayFair cipher with some suitable example.	**07**
	b.	Explain Brute Force Attack in detail with example.	**07**
		OR	
4.	**a.**	Encrypt the message good morning with the Hill Cipher with the key [9 4 5 7]	**07**
	b.	Explain rail fence technique using a suitable example.	**07**
5.	**a.**	What is the purpose of s-Boxes in DES? Explain the avalanche effect.	**07**
	b.	Explain the various modes of operation of black cipher. Explain any three of them.	**07**
		OR	
6.	**a.**	Explain the properties of Hash Function in details with example.	**07**
	b.	Write the Euclid's algorithm. List out the steps used to find the gcd of two numbers using Euclid algorithm.	**07**
7.	**a.**	What is RSA Algorithm? Why it is secure than MD5?	**10**
	b.	Explain public key infrastructure and public key certificate.	**04**
		OR	
8.	**a.**	Define Authentication. Explain the one way and two way authentication in X.509.	**07**
	b.	Explain the X.509 standard. What is the purpose of X.509 standard?	**07**